



Statistics of Financial Markets

Seminar 12: Scoring models



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Acronyms used in this material

- AI: Artificial Intelligence
- AR: Accuracy Ratio (of the CAP curve)
- AUC: Area Under the (ROC) Curve
- CAP: Cumulative Accuracy Profile
- CI: Confidence Interval
- DM: Deutsche Mark (the German mark)
- EAD: Exposure At Default
- FN: False Negative
- FPR: False Positive Rate
- HL: Hosmer–Lemeshow (test)
- IRB: Internal Ratings-Based (approach)
- KS: Kolmogorov–Smirnov (statistic)
- LDA: Linear Discriminant Analysis
- LGD: Loss Given Default
- PD: Probability of Default
- PDO: Points to Double the Odds
- ROC: Receiver Operating Characteristic (curve)
- RWA: Risk-Weighted Assets
- TN: True Negative
- TP: True Positive
- TPR: True Positive Rate
- UCI: University of California, Irvine (Machine Learning Repository)

Today's Question and Route

- **Question:** how does a bank turn the data of an applicant into a probability of default, and how do we judge that probability?
 - ▶ this seminar comes **before** Lecture 12: the section “What You Need for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: odds and log-odds, WoE and IV, the confusion matrix and AUC, scorecard points, on paper
 - ▶ Part B: a scorecard on real credit data, its validation, logit against LDA, capital and fairness, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	the PD of an applicant from a logit model; odds ratios	Solved, Proposed	A1
A3, A4	WoE and IV from a table of goods and bads	Solved, Proposed	A3
A5, A6	confusion matrix, AUC by counting pairs, Gini, KS, Brier score	Solved, Proposed	A5
A7, A8	scorecard scaling and points; prior correction and expected loss	Solved, Proposed	A7
B1, B2	IV and a WoE logit: South German Credit; IV of the Taiwan data	Solved, Proposed	B1
B3, B4	validation of the scorecard; logit against LDA on the Taiwan data	Solved, Proposed	B3
B5, B6	expected loss and IRB capital of a portfolio; fairness by sex and age	Solved, Proposed	B5
C1, C2	does gradient boosting beat the logit? what is wrong in an AI answer?	Proposed	B3, A5

- **[Solved]:** full solution in the slides and in the notebook, a model to follow; **[Proposed]:** you solve it, following the model

Data Used

Data set	Source	Size	Outcome
South German Credit	South German Credit (2020)	1000 credits, 1973–1975	300 bad, 700 good
Taiwan credit cards	Default of Credit Card Clients (2009)	30 000 card holders, 2005	default in October 2005 (22.1%)

- Both under CC BY 4.0, saved once in data/credit of the course repository; South German Credit is the corrected version of the UCI German credit data [Grömping \(2019\)](#)
- Default = 1 for a bad credit or a missed payment, 0 otherwise; bad credits are oversampled in South German Credit (population rate about 5%)
- In the notebook: `load_credit('sgc')`, `woe_table(x, y, var)`, `logit_fit(X, y)`, `auc(score, y)`; no account or key is needed

What You Need for Today (1/5): Default and Expected Loss

- **Default:** the borrower does not pay as agreed (Basel: more than 90 days past due, or unlikely to pay) [Basel Committee \(2006\)](#)
 - ▶ $Y = 1$ for default (a “bad”), $Y = 0$ for a good payer; a **credit score** orders applicants by risk
- **PD:** probability of default in one year; **LGD:** share of the exposure lost if default occurs; **EAD:** amount owed at default
 - ▶ **expected loss:** $EL = PD \times LGD \times EAD$
 - ▶ Basel IRB capital for other retail loans: $K = LGD[\Phi((\Phi^{-1}(PD) + \sqrt{R}\Phi^{-1}(0.999))/\sqrt{1-R}) - PD]$, $RWA = 12.5 K EAD$
 - ▶ R falls from 0.16 to 0.03 as PD grows; RWA: risk-weighted assets
- **Oversampling:** if the sample has a bad rate π_s and the population π , add $\ln \frac{\pi/(1-\pi)}{\pi_s/(1-\pi_s)}$ to the log-odds of every applicant
 - ▶ the ranking of the applicants does not change, the level of the PDs does

What You Need for Today (2/5): Odds and the Logit

- **Odds** of default $p/(1 - p)$; **log-odds** $\ln[p/(1 - p)]$; good:bad odds $(1 - p)/p$
 - ▶ $p = 0.2$: odds 0.25, log-odds -1.386 , good:bad odds 4:1
- **Logit model** [Cox \(1958\)](#): $\ln \frac{p}{1 - p} = \eta = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k, \quad p = \frac{1}{1 + e^{-\eta}}$
 - ▶ β_0 : the constant (intercept); estimated by maximum likelihood
 - ▶ β_j : change in the log-odds per unit of x_j ; e^{β_j} : **odds ratio**, the factor that multiplies the odds
 - ▶ effect on the probability: $\Delta p \approx p(1 - p)\beta_j$
- **LDA** (linear discriminant analysis, [Fisher \(1936\)](#)): $w = S_W^{-1}(m_1 - m_0)$ separates the groups best
 - ▶ m_1, m_0 : means of bads and goods; S_W : their common (pooled) covariance; with Normal groups the log-odds are linear, as in the logit

What You Need for Today (3/5): WoE and IV

- Split a variable into bins; g_i , b_i : goods and bads in bin i ; G , B : their totals
 - ▶ **Weight of Evidence:** $\text{WoE}_i = \ln \frac{g_i/G}{b_i/B}$; positive: safer than average, negative: riskier
- **Information Value:** $\text{IV} = \sum_i (g_i/G - b_i/B) \text{WoE}_i$
 - ▶ rule of thumb [Siddiqi \(2006\)](#): below 0.02 useless, 0.02–0.1 weak, 0.1–0.3 medium, above 0.3 strong
 - ▶ a bin without goods or without bads: add 0.5 to both counts
- A **WoE logit**: every variable is replaced by the WoE of its bin; bins and WoE come from the training sample only
 - ▶ the training sample is used for estimation; the test sample plays the future applicants and is used once, at the end

What You Need for Today (4/5): Validation

- Cut-off c : reject if $PD > c$. **TP**: bad rejected, **FN**: bad accepted, **FP**: good rejected, **TN**: good accepted
 - ▶ $TPR = TP / (TP + FN)$; $FPR = FP / (FP + TN)$; $accuracy = (TP + TN) / n$; $cost = (C_{FN} FN + C_{FP} FP) / n$
 - ▶ Bayes cut-off: reject if $p > C_{FP} / (C_{FP} + C_{FN})$; with $C_{FN} = 5$, $C_{FP} = 1$: $p > 1/6$
- **ROC curve**: (FPR, TPR) for all cut-offs; **AUC** = $P(\text{PD of a random bad} > \text{PD of a random good})$, ties count 1/2 [Hanley & McNeil \(1982\)](#)
 - ▶ by counting: the share of (bad, good) pairs in the right order; **Gini** = $2 \text{AUC} - 1$
 - ▶ **KS** = $\max_s |F_{\text{bad}}(s) - F_{\text{good}}(s)|$: the largest gap between the distribution functions of the PDs of bads and goods
- **Brier score** [Brier \(1950\)](#): $\frac{1}{n} \sum_i (p_i - y_i)^2$; **calibration**: the mean PD of a group should equal its default rate
 - ▶ Hosmer–Lemeshow [Hosmer & Lemeshow \(1980\)](#): $\sum_g (O_g - E_g)^2 / [E_g(1 - \bar{p}_g)] \approx \chi^2(G - 2)$; DeLong [DeLong et al. \(1988\)](#): standard error and test of AUC

What You Need for Today (5/5): Scorecards and Fairness

- ▣ **Scorecard:** $\text{Score} = \text{Offset} + \text{Factor} \times \ln(\text{good:bad odds})$ [Siddiqi \(2006\)](#)
 - ▶ **PDO** (points to double the odds): $\text{Factor} = \text{PDO} / \ln 2$; base score at odds_0 : $\text{Offset} = \text{Base} - \text{Factor} \ln(\text{odds}_0)$
 - ▶ points of attribute i of variable j (WoE logit with k variables): $-(\beta_j \text{WoE}_{ij} + \beta_0/k) \text{Factor} + \text{Offset}/k$
- ▣ **Fairness** at a cut-off [Hardt, Price & Srebro \(2016\)](#)
 - ▶ demographic parity: equal approval rates across groups
 - ▶ equal opportunity: equal approval rates of the good payers across groups
 - ▶ calibration within groups: the same PD means the same default rate in every group

A1: The PD of an Applicant [Solved]

Task

- ▣ Logit for default: constant -2.101 ; duration (years) 0.385 ; age (decades) -0.159 ; amount (1000 DM) 0.029 ; no checking account 1.859 ; balance < 0 DM 1.338 .
- ▣ Applicant: 2 years, 30 years old, 3000 DM, no checking account.
- ▣ 1. Compute the log-odds η and the PD.
- ▣ 2. Compute the odds of default and interpret $e^{1.859}$.
- ▣ 3. Compute the PD with one more year of duration and compare it with $p(1-p)\beta$.
- ▣ Report: four numbers and one sentence.

Solution

- ▣ 1.
 $\eta = -2.101 + 0.385 \times 2 - 0.159 \times 3 + 0.029 \times 3 + 1.859 = 0.138$;
 $PD = 1/(1 + e^{-0.138}) = 53.4\%$
- ▣ 2. odds $= e^{0.138} = 1.148$; $e^{1.859} = 6.42$: without a checking account the odds are 6.42 times those of the reference group
- ▣ 3. $\eta + 0.385$: PD = 62.8% (+9.3 points); $p(1-p)\beta = 9.6$ points
- ▣ The odds ratio is constant; the change in PD depends on where the applicant is.

A2: A Second Applicant [Proposed]

Solution discussed in class

Task

- ▣ The same logit as in A1. Applicant: 1 year, 45 years old, 2000 DM, balance below 0 DM. Model: A1.
- ▣ 1. Compute η , the odds and the PD.
- ▣ 2. Compute the PD with one more year of duration.
- ▣ 3. Compare the change in PD with that of A1 and explain the difference.
- ▣ Report: four numbers and one sentence.

A3: WoE and IV of the Checking Account [Solved]

Task

- ▣ South German Credit, full sample: goods and bads by checking account: no account 139/135; below 0 DM 164/105; 0–200 DM 49/14; at least 200 DM or salary 348/46.
- ▣ 1. Compute g_i/G and b_i/B for each group.
- ▣ 2. Compute the WoE of each group.
- ▣ 3. Compute the IV and classify the variable.
- ▣ Report: a table and one sentence.

Solution

	g	b	g/G	b/B	WoE	IV
no account	139	135	0.199	0.450	−0.818	0.206
< 0 DM	164	105	0.234	0.350	−0.401	0.046
0–200 DM	49	14	0.070	0.047	0.405	0.009
≥ 200 DM / salary	348	46	0.497	0.153	1.176	0.404
total	700	300	1	1		0.666

- ▣ e.g. $\ln(0.497/0.153) = 1.176$; $IV = 0.666 > 0.3$: strong.
- ▣ WoE falls monotonically from the safest to the riskiest group.

A4: WoE and IV of the Savings [Proposed]

Solution discussed in class

Task

- ▣ Goods/bads by savings: none or unknown 386/217; below 100 DM 69/34; 100–500 DM 52/11; 500–1000 DM 42/6; at least 1000 DM 151/32. Model: A3.
- ▣ 1. Compute the WoE of each group.
- ▣ 2. Compute the IV and classify the variable.
- ▣ 3. Propose a merger of groups with similar WoE.
- ▣ Report: a table and two sentences.

A5: Confusion Matrix and AUC by Counting [Solved]

Task

- ▣ Ten applicants, PD and outcome ($Y = 1$: default): 0.05 (0), 0.10 (0), 0.12 (0), 0.20 (0), 0.25 (1), 0.30 (0), 0.40 (0), 0.55 (1), 0.60 (0), 0.80 (1).
- ▣ 1. Build the confusion matrix at the cut-off 0.5 and compute accuracy, TPR and FPR.
- ▣ 2. Compute the AUC by counting the (bad, good) pairs in the right order, and the Gini.
- ▣ 3. Compute the KS statistic and the Brier score.
- ▣ Report: seven numbers and one sentence.

Solution

- ▣ 1. TP 2, FN 1, FP 1, TN 6: accuracy 80.0%, TPR 66.7%, FPR 14.3%
- ▣ 2. goods below each bad: $4 + 6 + 7 = 17$ of 21 pairs: AUC = 0.810, Gini = 0.619
- ▣ 3. at $PD \leq 0.20$: 4/7 of goods, 0/3 of bads: KS = 0.571; Brier = $1.4819/10 = 0.148$
- ▣ The bad with PD 0.25 is ranked below three goods and the bad with 0.55 below one: 4 of 21 pairs are in the wrong order.

A6: A Second Sample [Proposed]

Solution discussed in class

Task

- ▣ Eight applicants: 0.08 (0), 0.15 (0), 0.22 (1), 0.35 (0), 0.42 (0), 0.50 (1), 0.65 (1), 0.90 (0). Model: A5.
- ▣ 1. Build the confusion matrix at the cut-off 0.4 and compute accuracy, TPR and FPR.
- ▣ 2. Compute the AUC by counting pairs, and the Gini.
- ▣ 3. Compute the KS statistic and the Brier score.
- ▣ Report: seven numbers and one sentence.

A7: Scaling a Scorecard [Solved]

Task

- ▣ PDO = 20; 600 points at good:bad odds 50:1. A WoE logit with $k = 6$ variables has constant $\beta_0 = -0.850$ and, for the checking account, $\beta = -0.829$; WoE: no account -0.826 , at least 200 DM 1.043.
- ▣ 1. Compute the Factor and the Offset.
- ▣ 2. Compute the score of a PD of 5% and the PD of a score of 560.
- ▣ 3. Compute the points of the two checking-account groups.
- ▣ Report: six numbers and one sentence.

Solution

- ▣ 1. Factor = $20 / \ln 2 = 28.85$; Offset = $600 - 28.85 \ln 50 = 487.12$
- ▣ 2. $487.12 + 28.85 \times \ln 19 = 572.08$; 560: $\ln \text{odds} = 2.526$, odds 12.50:1, PD = 7.4%
- ▣ 3. Offset/ $k = 81.19$, $\beta_0/k = -0.1417$: no account $-(-0.829 \times (-0.826) + (-0.1417)) \times 28.85 + 81.19 = 65.5$; at least 200 DM: 110.2
- ▣ A well-funded account is worth 44.7 points more than no account: 2.2 doublings of the good:bad odds.

A8: Another Scale, Prior Correction and Expected Loss [Proposed]

Solution discussed in class

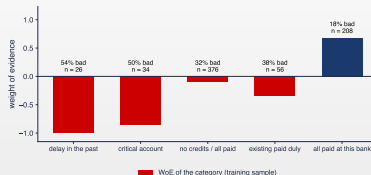
Task

- ▣ A bank uses $PDO = 40$ and 500 points at odds 20:1. A model built on a sample with 30% bads gives an applicant $PD = 30\%$; the population bad rate is 5%; $EAD = 10\,000$ DM, $LGD = 45\%$. Model: A7.
- ▣ 1. Compute the Factor, the Offset, the score of a PD of 10% and the PD of a score of 520.
- ▣ 2. Correct the PD of 30% to the population.
- ▣ 3. Compute the expected loss with both PDs.
- ▣ Report: seven numbers and one sentence.

B1: A WoE Scorecard for South German Credit [Solved]

- **Question:** which variables carry information about default, and how well does a WoE logit rank new applicants?
- **Inputs:** South German Credit; a stratified split: 700 credits for estimation, 300 for testing (30% bads in both)
- **Tasks**
 1. Compute the WoE tables and the IV of the 20 variables on the training sample (duration, amount and age in five fixed bins).
 2. Keep the variables with IV of at least 0.1 and estimate a logit for default on their WoE values.
 3. Compute the AUC on the training and on the test sample, with a 95% DeLong interval for the test AUC.
 4. Draw the WoE of the credit-history groups.
 5. Interpretation: does the order of the credit-history groups make economic sense?
- **Report:** the list of selected variables with their IV, the coefficients, two AUCs with the interval, the chart and two sentences
- **Notebook:** section B1

B1: Solution [Solved]



	constant	checking account	credit history	purpose	savings	duration	years with the employer
IV		0.591	0.215	0.205	0.166	0.159	0.122
$\hat{\beta}$	-0.850	-0.829	-0.675	-0.937	-0.726	-1.082	-0.873
se	0.095	0.127	0.206	0.217	0.246	0.239	0.265

- 6 variables with $IV \geq 0.1$; AUC: training 0.789, test 0.801, 95% CI [0.749, 0.854]
- Interpretation: yes: a delay in the past (54% bad) and a critical account (50%) are the riskiest, all credits at this bank paid back duly (18%) the safest; in the uncorrected UCI labels this safest group was called "critical account"

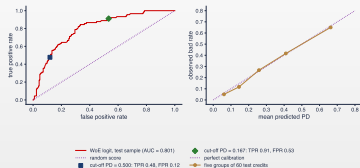
B2: Information Value of the Taiwan Data [Proposed]

- **Question:** which kind of information predicts the default of a card holder: payment behaviour or personal data? Model: B1.
- **Inputs:** Taiwan credit cards, 30 000 clients; the same stratified 70/30 split
- **Tasks**
 1. Compute the IV of PAY_0 and PAY_2 (repayment status in September and August; categories -2 to 3 or more), of LIMIT_BAL, PAY_AMT1, BILL_AMT1 and AGE (quintiles) and of SEX, EDUCATION and MARRIAGE (categories) on the training sample.
 2. Compute the default rate of each PAY_0 category.
 3. Rank the variables by IV and classify them with the rule of thumb.
 4. Interpretation: why is the last month's repayment status so much stronger than the personal data?
- **Report:** a table of nine IVs, six default rates and two sentences
- **Notebook:** section B2

B3: Validating the Scorecard [Solved]

- ▣ **Question:** how good is the B1 scorecard on the test sample, and which cut-off should the bank use?
- ▣ **Inputs:** the 300 test credits of B1 with their PDs; costs: 5 for an accepted bad, 1 for a rejected good
- ▣ **Tasks**
 1. Build the confusion matrix at the cut-offs 0.5 and 1/6, with accuracy, TPR, FPR and the cost per applicant.
 2. Compute the AUC, the Gini, the accuracy ratio of the CAP curve and the KS statistic.
 3. Compute the Brier score, compare it with a constant PD of 30%, and run the Hosmer–Lemeshow test with five groups.
 4. Draw the ROC curve with the two cut-offs and the calibration plot.
 5. Interpretation: which cut-off would you recommend to the bank?
- ▣ **Report:** a table of two confusion matrices, six statistics, the chart and two sentences
- ▣ **Notebook:** section B3

B3: Solution [Solved]



cut-off	TP	FN	FP	TN	accuracy	TPR	FPR	cost
0.5	43	47	25	185	76.0%	47.8%	11.9%	0.867
1/6	82	8	112	98	60.0%	91.1%	53.3%	0.507

- AUC = 0.801, Gini = 0.603, AR = 0.602, KS = 0.511; Brier = 0.162 against 0.210; HL = 0.589 ($p = 0.899$): no evidence of miscalibration
- Interpretation: the cut-off of 1/6: it rejects more applicants (64.7%) but catches 91.1% of the bads, and the cost falls from 0.867 to 0.507; accuracy is the wrong criterion when errors have different costs

B4: Logit or LDA on the Taiwan Data? [Proposed]

- **Question:** do the logit and Fisher's LDA differ in ranking or in calibration? Model: B3.
- **Inputs:** Taiwan, 70/30 split; 13 standardised inputs: log limit, age, indicators of the September repayment status, number of late months, utilisation, log payments
- **Tasks**
 1. Estimate a logit and an LDA on the training sample with the same inputs.
 2. Compute the test AUC of both and the DeLong test of equal AUCs.
 3. Compute the Brier score, the Hosmer–Lemeshow statistic (ten groups) and the mean PD of both, against the default rate.
 4. Compute the accuracy of the rule “nobody defaults” and of both models at the cut-off 0.5.
 5. Interpretation: is the LDA worse at ranking or at calibration?
- **Report:** one table and two sentences
- **Notebook:** section B4

B5: From PDs to Capital [Solved]

- **Question:** how much expected loss and Basel capital does the test portfolio of B1 carry, and how much does the prior correction matter?
- **Inputs:** the 300 test credits of B1; $EAD = \text{credit amount}$; $LGD = 45\%$; population bad rate 5%; other retail exposures
- **Tasks**
 1. Correct the PD of every credit from the sample rate of 30% to the population rate of 5%.
 2. Compute the expected loss of the portfolio, in DM and as a share of EAD, with and without the correction.
 3. Compute the IRB capital K and the risk-weighted assets of the portfolio with the corrected PDs.
 4. Interpretation: why does the correction change the expected loss but not the AUC?
- **Report:** eight numbers and two sentences
- **Notebook:** section B5

B5: Solution [Solved]

- Shift of the log-odds: $\ln(0.05/0.95) - \ln(0.30/0.70) = -2.097$
 - ▶ mean PD: 30.7% in the sample scale, 7.3% after the correction (median 4.2%, maximum 54.4%)
- EAD 957 819 DM; EL 41 744 DM = 4.4% of EAD (without correction: 16.3%)
 - ▶ IRB capital K : 54 542 DM = 5.7% of EAD; RWA 681 771 DM, a risk weight of 71.2%
- The mean corrected PD is above 5% because the correction is applied to the log-odds, not to the PD
- Interpretation: the correction adds the same constant to every log-odds: the order of the applicants, hence AUC, is unchanged, while every PD shrinks and EL falls from 16.3% to 4.4% of EAD

B6: Fairness on the Taiwan Data [Proposed]

- **Question:** does the logit of B4 treat men and women, and young and older clients, alike? Model: B5 and slide 5/5.
- **Inputs:** Taiwan test sample (9000 clients); sex and age are not model inputs; approve the 75% with the lowest PD
- **Tasks**
 1. Compute the approval rate and the default rate of men and women, and of clients up to 25 years and older.
 2. Compute the approval rate of the good payers (equal opportunity) in each group.
 3. Compute the default rate among the approved clients of each group.
 4. Interpretation: which fairness criterion does the model come closest to satisfying?
- **Report:** one table and two sentences
- **Notebook:** section B6

C1: Does Gradient Boosting Beat the Logit? [Proposed]

- ▣ **Question:** benchmarks report gains for flexible models [Lessmann et al. \(2015\)](#); how large is the gain on the Taiwan data, and is it worth the loss of a simple scorecard?
- ▣ **Inputs:** the inputs and the split of B4; models: B3, B4
- ▣ **Tasks**
 1. Fit gradient boosting (scikit-learn, default settings) on the training sample.
 2. Compare its test AUC with the logit by the DeLong test, and compare the Brier scores.
 3. Compare the training and the test AUC of gradient boosting.
 4. Interpretation: is the gain large enough to replace the scorecard?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the B1 scorecard. The answer:
 - (a) The AUC is 0.80, so the scorecard classifies 0.80 of the applicants correctly.
 - (b) The Gini coefficient is $\text{AUC} - 0.5 = 0.30$.
 - (c) An applicant with a PD of 30% from this model has a 30% chance of default at the bank.
 - (d) With $\text{PDO} = 20$, a PD that halves from 20% to 10% adds exactly 20 points.
 - (e) KS is the largest vertical gap between the score distributions of bads and goods.
 - (f) LDA may only be used when every variable is Normally distributed.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ A logit models the log-odds; e^{β} is an odds ratio; the effect on PD is $p(1 - p)\beta$
- ▣ WoE and IV turn a table of goods and bads into evidence and a ranking of variables
- ▣ AUC counts correctly ordered pairs; $\text{Gini} = 2 \text{AUC} - 1$; accuracy misleads when classes are imbalanced
- ▣ Choose the cut-off from the costs; correct oversampled PDs before any EL or capital
- ▣ An AI answer is a draft: check what AUC means, the Gini formula, the prior and the PDO scale

After the Seminar

- Lecture 12 develops each topic of today: the Basel parameters, WoE and IV, logit and LDA, scorecards, validation, reject inference, fairness, Altman and Merton
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: cross-validated benchmark, costs and fairness by group
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 21–22; [Härdle & Simar \(2019\)](#), Ch. 14; [Siddiqi \(2006\)](#)

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